

Analysing the final Mapper Graph

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```
scaler <- "zscale"

# Min-Max Normalisation (commonly used in Kepler Mapper)
min_max_norm <- function(x) {
  (x - min(x)) / (max(x) - min(x))
}

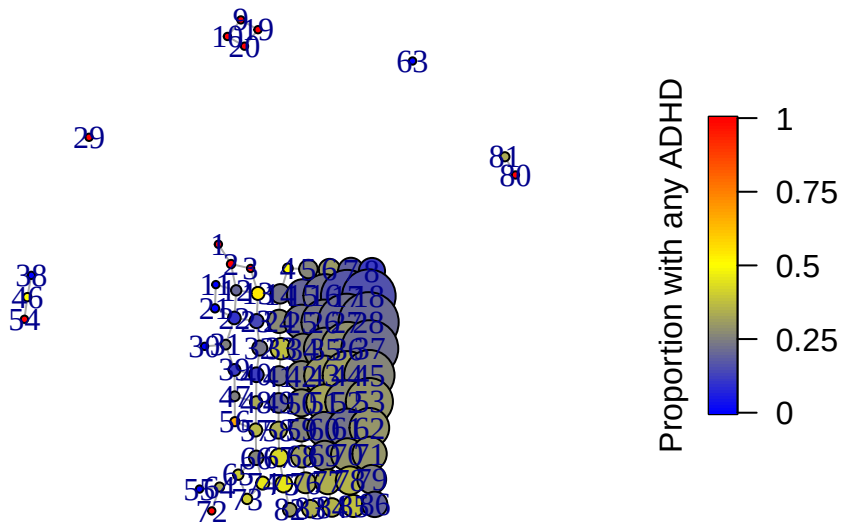
if (scaler == "minmax") {
  mapper_scaled <- mapper_data |>
    mutate(across(everything(), min_max_norm))
} else if (scaler == "zscale") {
  mapper_scaled <- mapper_data |>
    mutate(across(everything(), ~ scale(.)[, 1]))
} else {
  stop("No valid scaler selected")
}

mapper_pca <- prcomp(mapper_data, center = TRUE, scale = TRUE)
```

```
# 10 intervals, 30% overlap
# DBSCAN(0.4, 26)
mapper <- MapperAlgo(
  original_data = mapper_scaled,
  filter_values = mapper_pca$x[, 1:2],
  percent_overlap = 30,
  methods = "dbscan",
  method_params = list(eps = 0.4, minPts = 26),
  cover_type = "stride",
  intervals = 10,
  num_cores = 12
)
# apply index fix
```

```
# mapper$points_in_vertex <- lapply(mapper$points_in_vertex, function(x) x - 1)

create_mapper_graph(
  mapper_obj = mapper,
  colourisation = any_adhd,
  original_data = mapper_data,
  graph_layout = layout_with_kk,
  legend = TRUE,
  legend_title = "Proportion with any ADHD",
  legend_tick_num = 5
)
```



```
IGRAPH 7e59b9b U--- 86 131 --
+ attr: size (v/n), color (v/c)
+ edges from 7e59b9b:
  [1] 1-- 2 2-- 3 2--12 3--13 4-- 5 4--14 5-- 6 5--15 6-- 7 6--16
 [11] 7-- 8 7--17 8--18 9--10 9--19 10--20 11--12 11--21 12--13 12--22
 [21] 13--14 13--23 14--15 14--24 15--16 15--25 16--17 16--26 17--18 17--27
 [31] 18--28 19--20 21--22 22--23 22--31 23--24 23--32 24--25 24--33 25--26
 [41] 25--34 26--27 26--35 27--28 27--36 28--37 30--31 31--39 32--33 32--40
 [51] 33--34 33--41 34--35 34--42 35--36 35--43 36--37 36--44 37--45 38--46
 [61] 39--40 39--47 40--41 40--48 41--42 41--49 42--43 42--50 43--44 43--51
 [71] 44--45 44--52 45--53 46--54 47--48 47--56 48--49 48--57 49--50 49--58
+ ... omitted several edges
```

Analysing the isolated components

We shall analyse the smaller isolated components:

```
isolated_components <- find_isolated_components(mapper_graph_data, mapper_full_data)
```

6 number of components

Found 5 isolated component(s) detached from the main trunk.

```
interested_cols <- c(
  "pid", "wave", "adhd",
  "wemwbs_total", "life_sat", "aff_val",
  "bangs_auto_sat", "bangs_auto_frus", "bangs_comp_sat",
  "bangs_comp_frus", "bangs_rel_sat", "bangs_rel_frus",
  "gaming_value_sum", "total_wave_minutes", "max_binge_minutes",
  "binge_day_of_week", "mean_mobile", "mean_nintendo",
  "mean_steam", "mean_xbox", "score_mean", "atten_score_mean"
)

# majority component is 1
for (comp in unique(isolated_components$component)) {
  if (comp == 1) next

  print(paste0("Component ", comp, ":\n"))
  comp_nodes <- isolated_components |>
    filter(component == comp) |>
    pull(node)
  n_nodes <- length(comp_nodes)
  unique_rows <- unique(unlist(
    mapper_graph_data$points_in_vertex[c(comp_nodes)]
  ))
  comp_data <- mapper_full_data[unique_rows, ]
  print(paste0(
    "Nodes (",
    n_nodes,
    ") Rows of data (",
    length(unique_rows), ")"
  ))
  print(skimr::skim(comp_data[, interested_cols]))
}
```

```

[1] "Component 2:\n"
[1] "Nodes (4) Rows of data (1)"
-- Data Summary -----

```

	Values
Name	comp_data[, interested_co...
Number of rows	1
Number of columns	22

```

-----
Column type frequency:
  character      1
  factor         2
  numeric       19
-----
Group variables      None

```

```

-- Variable type: character -----
  skim_variable n_missing complete_rate min max empty n_unique whitespace
1 pid          0          1  11  11    0          1          0

```

```

-- Variable type: factor -----
  skim_variable      n_missing complete_rate ordered n_unique
1 adhd              0          1 FALSE          1
2 binge_day_of_week  0          1 FALSE          1
  top_counts
1 Neu: 1, Dia: 0, Sel: 0
2 Tue: 1, Mon: 0, Wed: 0, Thu: 0

```

```

-- Variable type: numeric -----
  skim_variable      n_missing complete_rate mean sd    p0    p25    p50    p75
1 wave          0          1    1    NA    1    1    1    1
2 wemwbs_total    0          1   22    NA   22   22   22   22
3 life_sat        0          1    5    NA    5    5    5    5
4 aff_val         0          1   75    NA   75   75   75   75
5 bangs_auto_sat  0          1   13    NA   13   13   13   13
6 bangs_auto_frus 0          1   14    NA   14   14   14   14
7 bangs_comp_sat  0          1   17    NA   17   17   17   17
8 bangs_comp_frus 0          1   11    NA   11   11   11   11
9 bangs_rel_sat   0          1   17    NA   17   17   17   17
10 bangs_rel_frus 0          1   10    NA   10   10   10   10
11 gaming_value_sum 0          1   26    NA   26   26   26   26
12 total_wave_minutes 0          1  234    NA  234  234  234  234
13 max_binge_minutes 0          1  154    NA  154  154  154  154
14 mean_mobile     0          1    0    NA    0    0    0    0

```

15 mean_nintendo	0	1	0	NA	0	0	0	0
16 mean_steam	0	1	15.6	NA	15.6	15.6	15.6	15.6
17 mean_xbox	0	1	0	NA	0	0	0	0
18 score_mean	0	1	45	NA	45	45	45	45
19 atten_score_mean	0	1	45	NA	45	45	45	45

p100 hist

1	1
2	22
3	5
4	75
5	13
6	14
7	17
8	11
9	17
10	10
11	26
12	234
13	154
14	0
15	0
16	15.6
17	0
18	45
19	45

[1] "Component 3:\n"

[1] "Nodes (1) Rows of data (1)"

-- Data Summary -----

	Values
Name	comp_data[, interested_co...
Number of rows	1
Number of columns	22

Column type frequency:

character	1
factor	2
numeric	19

Group variables None

-- Variable type: character -----

	skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
1	pid	0	1	11	11	0	1	0

```
-- Variable type: factor -----
skim_variable      n_missing complete_rate ordered n_unique
1 adhd              0              1 FALSE          1
2 binge_day_of_week 0              1 FALSE          1
top_counts
1 Dia: 1, Neu: 0, Sel: 0
2 Thu: 1, Mon: 0, Tue: 0, Wed: 0
```

```
-- Variable type: numeric -----
skim_variable      n_missing complete_rate   mean sd      p0      p25      p50
1 wave              0              1      2   NA      2       2       2
2 wemwbs_total      0              1     12   NA     12      12      12
3 life_sat          0              1      2   NA      2       2       2
4 aff_val            0              1     40   NA     40      40      40
5 bangs_auto_sat    0              1     15   NA     15      15      15
6 bangs_auto_frus   0              1     20   NA     20      20      20
7 bangs_comp_sat    0              1     20   NA     20      20      20
8 bangs_comp_frus   0              1     13   NA     13      13      13
9 bangs_rel_sat     0              1     16   NA     16      16      16
10 bangs_rel_frus   0              1     21   NA     21      21      21
11 gaming_value_sum 0              1     27   NA     27      27      27
12 total_wave_minutes 0              1 1747.   NA 1747.  1747.  1747.
13 max_binge_minutes 0              1   522.   NA 522.  522.  522.
14 mean_mobile       0              1      0   NA      0       0       0
15 mean_nintendo     0              1    2.10  NA    2.10  2.10  2.10
16 mean_steam        0              1   114.   NA  114.  114.  114.
17 mean_xbox         0              1      0   NA      0       0       0
18 score_mean        0              1     68   NA     68      68      68
19 atten_score_mean 0              1     68   NA     68      68      68
      p75    p100 hist
1      2      2
2     12     12
3      2      2
4     40     40
5     15     15
6     20     20
7     20     20
8     13     13
9     16     16
10    21     21
11    27     27
12 1747.  1747.
```

```

13  522.    522.
14   0      0
15   2.10   2.10
16  114.    114.
17   0      0
18   68     68
19   68     68
[1] "Component 4:\n"
[1] "Nodes (3) Rows of data (2)"
-- Data Summary -----

```

	Values
Name	comp_data[, interested_co...
Number of rows	2
Number of columns	22

```

-----
Column type frequency:
  character      1
   factor       2
  numeric      19
-----
Group variables      None

```

```

-- Variable type: character -----
  skim_variable n_missing complete_rate min max empty n_unique whitespace
1 pid           0           1  11  11    0      2           0

```

```

-- Variable type: factor -----
  skim_variable      n_missing complete_rate ordered n_unique
1 adhd              0           1 FALSE      2
2 binge_day_of_week  0           1 FALSE      2
  top_counts
1 Dia: 1, Neu: 1, Sel: 0
2 Thu: 1, Fri: 1, Mon: 0, Tue: 0

```

```

-- Variable type: numeric -----
  skim_variable      n_missing complete_rate      mean      sd      p0      p25
1 wave            0           1      2.5      0.707      2      2.25
2 wemwbs_total    0           1     15      7.07     10     12.5
3 life_sat        0           1      1.5      2.12      0      0.75
4 aff_val         0           1     22.5     24.7      5     13.8
5 bangs_auto_sat  0           1      7.5      4.95      4      5.75
6 bangs_auto_frus 0           1     17      5.66     13     15
7 bangs_comp_sat  0           1      6       4.24      3      4.5

```

8	bangs_comp_frus	0	1	13.5	3.54	11	12.2
9	bangs_rel_sat	0	1	6.5	4.95	3	4.75
10	bangs_rel_frus	0	1	19.5	2.12	18	18.8
11	gaming_value_sum	0	1	17.5	3.54	15	16.2
12	total_wave_minutes	0	1	2742.	3203.	477.	1610.
13	max_binge_minutes	0	1	444.	253.	264.	354.
14	mean_mobile	0	1	0	0	0	0
15	mean_nintendo	0	1	0.137	0.193	0	0.0683
16	mean_steam	0	1	183.	214.	31.6	107.
17	mean_xbox	0	1	0	0	0	0
18	score_mean	0	1	65.2	4.01	62.3	63.8
19	atten_score_mean	0	1	65.2	4.01	62.3	63.8

	p50	p75	p100	hist
1	2.5	2.75	3	
2	15	17.5	20	
3	1.5	2.25	3	
4	22.5	31.2	40	
5	7.5	9.25	11	
6	17	19	21	
7	6	7.5	9	
8	13.5	14.8	16	
9	6.5	8.25	10	
10	19.5	20.2	21	
11	17.5	18.8	20	
12	2742.	3875.	5007.	
13	444.	533.	623.	
14	0	0	0	
15	0.137	0.205	0.273	
16	183.	258.	334.	
17	0	0	0	
18	65.2	66.6	68	
19	65.2	66.6	68	

[1] "Component 5:\n"

[1] "Nodes (1) Rows of data (1)"

-- Data Summary -----

	Values
Name	comp_data[, interested_co...
Number of rows	1
Number of columns	22

Column type frequency:

character	1
factor	2


```

numeric                                19
-----
Group variables                        None

-- Variable type: character -----
skim_variable n_missing complete_rate min max empty n_unique whitespace
1 pid          0           1 11 11      0          1          0

-- Variable type: factor -----
skim_variable      n_missing complete_rate ordered n_unique
1 adhd              0           1 FALSE          1
2 binge_day_of_week 0           1 FALSE          1
top_counts
1 Neu: 1, Dia: 0, Sel: 0
2 Sat: 1, Mon: 0, Tue: 0, Wed: 0

-- Variable type: numeric -----
skim_variable      n_missing complete_rate mean sd      p0      p25      p50      p75
1 wave              0           1 1 NA      1      1      1      1
2 wemwbs_total      0           1 27 NA     27     27     27     27
3 life_sat          0           1 2 NA      2      2      2      2
4 aff_val           0           1 20 NA     20     20     20     20
5 bangs_auto_sat    0           1 3 NA      3      3      3      3
6 bangs_auto_frus   0           1 21 NA     21     21     21     21
7 bangs_comp_sat    0           1 3 NA      3      3      3      3
8 bangs_comp_frus   0           1 9 NA      9      9      9      9
9 bangs_rel_sat     0           1 3 NA      3      3      3      3
10 bangs_rel_frus   0           1 9 NA      9      9      9      9
11 gaming_value_sum 0           1 20 NA     20     20     20     20
12 total_wave_minutes 0           1 687. NA    687. 687. 687. 687.
13 max_binge_minutes 0           1 686. NA    686. 686. 686. 686.
14 mean_mobile      0           1 0 NA      0      0      0      0
15 mean_nintendo    0           1 0 NA      0      0      0      0
16 mean_steam       0           1 45.8 NA    45.8 45.8 45.8 45.8
17 mean_xbox        0           1 0 NA      0      0      0      0
18 score_mean       0           1 54 NA     54     54     54     54
19 atten_score_mean 0           1 54 NA     54     54     54     54
p100 hist
1 1
2 27
3 2
4 20
5 3

```

```

6 21
7 3
8 9
9 3
10 9
11 20
12 687.
13 686.
14 0
15 0
16 45.8
17 0
18 54
19 54
[1] "Component 6:\n"
[1] "Nodes (2) Rows of data (3)"
-- Data Summary -----

```

Name	Values
comp_data[, interested_co...	
Number of rows	3
Number of columns	22

```

-----
Column type frequency:
  character      1
  factor         2
  numeric       19
-----
Group variables      None

```

```

-- Variable type: character -----
skim_variable n_missing complete_rate min max empty n_unique whitespace
1 pid          0              1 11 11    0          3          0

```

```

-- Variable type: factor -----
skim_variable      n_missing complete_rate ordered n_unique
1 adhd              0              1 FALSE          2
2 binge_day_of_week 0              1 FALSE          3
top_counts
1 Neu: 2, Sel: 1, Dia: 0
2 Thu: 1, Sat: 1, Sun: 1, Mon: 0

```

```

-- Variable type: numeric -----
skim_variable      n_missing complete_rate      mean      sd      p0      p25

```

1	wave	0	1	3.33	2.31	2	2
2	wemwbs_total	0	1	19.3	3.51	16	17.5
3	life_sat	0	1	4.33	0.577	4	4
4	aff_val	0	1	59.7	20.0	40	49.5
5	bangs_auto_sat	0	1	14.3	4.62	9	13
6	bangs_auto_frus	0	1	13	1.73	12	12
7	bangs_comp_sat	0	1	14	2.65	11	13
8	bangs_comp_frus	0	1	9.67	1.53	8	9
9	bangs_rel_sat	0	1	10	4.58	5	8
10	bangs_rel_frus	0	1	11.3	5.51	6	8.5
11	gaming_value_sum	0	1	18.3	4.73	13	16.5
12	total_wave_minutes	0	1	1383.	738.	532.	1159.
13	max_binge_minutes	0	1	403.	103.	326.	345.
14	mean_mobile	0	1	0	0	0	0
15	mean_nintendo	0	1	0	0	0	0
16	mean_steam	0	1	92.2	49.2	35.4	77.3
17	mean_xbox	0	1	0	0	0	0
18	score_mean	0	1	59.3	4.18	55	57.3
19	atten_score_mean	0	1	59.3	4.18	55	57.3
	p50	p75	p100	hist			
1	2	4	6				
2	19	21	23				
3	4	4.5	5				
4	59	69.5	80				
5	17	17	17				
6	12	13.5	15				
7	15	15.5	16				
8	10	10.5	11				
9	11	12.5	14				
10	11	14	17				
11	20	21	22				
12	1787.	1809.	1830.				
13	364.	442.	520				
14	0	0	0				
15	0	0	0				
16	119.	121.	122.				
17	0	0	0				
18	59.7	61.5	63.3				
19	59.7	61.5	63.3				

```
print("Full data:")
```

```
[1] "Full data:"
```

```
print(skimr::skim(mapper_full_data))
```

```
-- Data Summary -----
```

	Values
Name	mapper_full_data
Number of rows	5836
Number of columns	157

```
-----  
Column type frequency:
```

character	59
factor	2
logical	9
numeric	85
POSIXct	2

```
-----  
Group variables      None
```

```
-- Variable type: character -----
```

skim_variable	n_missing	complete_rate	min	max	empty	n_unique
1 pid	0	1	11	11	0	1395
2 country	0	1	2	2	0	2
3 geo_area	0	1	2	4	0	874
4 panel_source	0	1	8	8	0	1
5 cohort	0	1	17	17	0	1
6 local_timezone	0	1	8	8	0	7
7 gender	0	1	3	17	0	4
8 ethnicity	0	1	5	42	0	9
9 edu_level	0	1	11	30	0	7
10 employment	0	1	5	34	0	7
11 marital_status	0	1	7	40	0	6
12 political_party	0	1	5	33	0	14
13 used_platforms	0	1	4	177	0	95
14 eligible_platforms	0	1	4	21	0	7
15 linked_platforms	82	0.986	3	30	0	20
16 nintendo_uses_family	3811	0.347	2	3	0	2
17 android_uses_family	5836	0	NA	NA	0	0
18 ios_uses_family	5836	0	NA	NA	0	0

19	steam_uses_family	784	0.866	2	3	0	2
20	xbox_uses_family	4947	0.152	2	3	0	2
21	neuro_identify	0	1	2	17	0	3
22	neuro_diagnosed	3554	0.391	2	17	0	3
23	neuro_diag_asd	5232	0.103	30	30	0	1
24	neuro_diag_adhd	4963	0.150	47	47	0	1
25	neuro_diag_dyslexia	5762	0.0127	8	8	0	1
26	neuro_diag_dyspraxia	5782	0.00925	9	9	0	1
27	neuro_diag_dyscalculia	5802	0.00583	11	11	0	1
28	neuro_diag_tourette	5830	0.00103	17	17	0	1
29	neuro_diag OCD	5642	0.0332	35	35	0	1
30	neuro_diag_spd	5784	0.00891	33	33	0	1
31	neuro_diag_nvld	5836	0	NA	NA	0	0
32	neuro_diag_specific_ld	5787	0.00840	62	62	0	1
33	neuro_diag_prefer_not_say	5816	0.00343	17	17	0	1
34	neuro_diag_other	5697	0.0238	4	132	0	30
35	neuro_iden_asd	5373	0.0793	30	30	0	1
36	neuro_iden_adhd	5342	0.0846	47	47	0	1
37	neuro_iden_dyslexia	5779	0.00977	8	8	0	1
38	neuro_iden_dyspraxia	5824	0.00206	9	9	0	1
39	neuro_iden_dyscalculia	5836	0	NA	NA	0	0
40	neuro_iden_OCD	5754	0.0141	35	35	0	1
41	neuro_iden_spd	5816	0.00343	33	33	0	1
42	neuro_iden_specific_ld	5828	0.00137	62	62	0	1
43	neuro_iden_prefer_not_say	5813	0.00394	17	17	0	1
44	neuro_iden_other	5810	0.00446	7	39	0	5
45	care_children	5055	0.134	24	24	0	1
46	care_elderly	5587	0.0427	40	40	0	1
47	care_disability	5585	0.0430	68	68	0	1
48	care_none	1312	0.775	56	56	0	1
49	care_prefer_not_say	5794	0.00720	17	17	0	1
50	care_other	5770	0.0113	4	238	0	16
51	dependents	42	0.993	1	17	0	6
52	geo_area_original	0	1	2	4	0	898
53	gender_original	0	1	3	21	0	16
54	edu_level_original	0	1	16	56	0	7
55	ethnicity_original	2	1.000	5	48	0	13
56	marital_status_original	0	1	6	75	0	27
57	political_party_original	0	1	4	47	0	42
58	employment_original	3	0.999	5	137	0	47
59	dependents_original	4566	0.218	1	17	0	6
	whitespace						
1	0						

2	0
3	0
4	0
5	0
6	0
7	0
8	0
9	0
10	0
11	0
12	0
13	0
14	0
15	0
16	0
17	0
18	0
19	0
20	0
21	0
22	0
23	0
24	0
25	0
26	0
27	0
28	0
29	0
30	0
31	0
32	0
33	0
34	0
35	0
36	0
37	0
38	0
39	0
40	0
41	0
42	0
43	0
44	0

45	0
46	0
47	0
48	0
49	0
50	0
51	0
52	0
53	0
54	0
55	0
56	0
57	0
58	0
59	0

-- Variable type: factor -----

	skim_variable	n_missing	complete_rate	ordered	n_unique
1	adhd	0	1	FALSE	3
2	binge_day_of_week	0	1	FALSE	8

top_counts

1 Neu: 4469, Dia: 873, Sel: 494

2 Sun: 964, Sat: 937, Fri: 741, Non: 696

-- Variable type: logical -----

	skim_variable	n_missing	complete_rate	mean	count
1	qualified	0	1	1	"TRU: 5836"
2	plays_xbox	0	1	0.220	"FAL: 4553, TRU: 1283"
3	plays_steam	0	1	0.866	"TRU: 5052, FAL: 784"
4	plays_nintendo	0	1	0.349	"FAL: 3800, TRU: 2036"
5	plays_ios	0	1	0.198	"FAL: 4680, TRU: 1156"
6	plays_android	0	1	0.238	"FAL: 4448, TRU: 1388"
7	plays_playstation	0	1	0.218	"FAL: 4561, TRU: 1275"
8	neuro_iden_tourette	5836	0	NaN	": "
9	neuro_iden_nvld	5836	0	NaN	": "

-- Variable type: numeric -----

	skim_variable	n_missing	complete_rate	mean	sd
1	wave	0	1	3.11	1.69
2	aff_val	0	1	63.4	21.3
3	wemwbs_total	0	1	23.2	5.26
4	bangs_auto_sat	0	1	17.1	2.90
5	bangs_auto_frus	0	1	7.61	3.53

6	bangs_comp_sat	0	1	16.1	3.07
7	bangs_comp_frus	0	1	7.95	3.89
8	bangs_rel_sat	0	1	12.1	4.69
9	bangs_rel_frus	0	1	6.43	3.85
10	gaming_value_sum	0	1	21.5	3.71
11	total_wave_minutes	0	1	1604.	1731.
12	max_binge_minutes	0	1	319.	248.
13	atten_score_mean	0	1	58.1	8.53
14	affective_valence	0	1	63.4	21.3
15	life_sat	0	1	5.12	2.00
16	bangs_1	0	1	5.69	1.13
17	bangs_2	0	1	5.78	1.11
18	bangs_3	0	1	5.61	1.17
19	bangs_4	0	1	2.57	1.47
20	bangs_5	0	1	2.56	1.42
21	bangs_6	0	1	2.48	1.49
22	bangs_7	0	1	5.20	1.25
23	bangs_8	0	1	5.60	1.17
24	bangs_9	0	1	5.32	1.23
25	bangs_10	0	1	2.64	1.55
26	bangs_11	0	1	2.70	1.57
27	bangs_12	0	1	2.61	1.50
28	bangs_13	0	1	4.02	1.80
29	bangs_14	0	1	4.31	1.79
30	bangs_15	0	1	3.78	1.73
31	bangs_16	0	1	2.23	1.55
32	bangs_17	0	1	2.11	1.42
33	bangs_18	0	1	2.10	1.47
34	gaming_value_work	0	1	3.85	0.873
35	gaming_value_social	0	1	4.68	1.31
36	gaming_value_cognitive	0	1	4.64	1.06
37	gaming_value_emotion	0	1	4.69	1.18
38	gaming_value_routines	0	1	3.60	1.12
39	wemwbs_1	0	1	3.06	1.07
40	wemwbs_2	0	1	3.17	0.935
41	wemwbs_3	0	1	3.54	0.897
42	wemwbs_4	0	1	3.34	0.989
43	wemwbs_5	0	1	3.66	0.928
44	wemwbs_6	0	1	3.16	1.01
45	wemwbs_7	0	1	3.31	0.949
46	days_played	0	1	7.78	4.92
47	mean_daily_minutes	0	1	107.	115.
48	median_daily_minutes	0	1	83.3	124.

49	sd_daily_minutes	0	1	98.9	77.5
50	cv_daily_minutes	0	1	1.31	0.985
51	mean_weekday_minutes	0	1	102.	115.
52	mean_weekend_minutes	0	1	120.	139.
53	mean_mobile	0	1	3.70	19.5
54	mean_steam	0	1	87.9	109.
55	mean_nintendo	0	1	2.33	14.0
56	mean_xbox	0	1	12.9	57.4
57	platforms_used	0	1	1.03	0.532
58	simon_waves_count	0	1	2.35	0.725
59	rt_mean	0	1	990.	229.
60	rt_sd	0	1	85.9	105.
61	rt_roc	0	1	-14.0	65.6
62	score_mean	0	1	58.1	8.53
63	survey_duration	0	1	694.	522.
64	diaries_completed	0	1	13.4	13.1
65	panels_completed	0	1	5.14	1.29
66	age	0	1	27.7	5.30
67	height	0	1	174.	10.1
68	weight	0	1	84.3	24.5
69	self_reported_weekly_play	564	0.903	1095.	693.
70	prop_steam	784	0.866	75.3	27.9
71	prop_xbox	4553	0.220	42.6	34.8
72	prop_playstation	4561	0.218	22.1	19.3
73	prop_nintendo	3811	0.347	30.9	31.1
74	prop_ios	4680	0.198	17.5	16.2
75	prop_android	4448	0.238	20.2	17.4
76	prop_other	5607	0.0392	39.9	24.8
77	total_recent_play_accounted_for	0	1	58.0	41.1
78	total_play_possible_to_collect	0	1	80.5	24.9
79	nintendo_first_party_prop	4284	0.266	0.615	0.301
80	dependents_imputed	0	1	0.775	0.417
81	height_original	413	0.929	174.	9.98
82	weight_original	413	0.929	84.3	24.7
83	height_imputed	0	1	0.0708	0.256
84	weight_imputed	0	1	0.0708	0.256
85	any_adhd	0	1	0.234	0.424

	p0	p25	p50	p75	p100	hist
1	1	2	3	5	7	
2	0	50	66	80	100	
3	7	20	23	27	35	
4	3	15	17	19	21	
5	3	5	7	10	21	

6	3	14	16	18	21
7	3	5	7	10	21
8	3	9	12	15	21
9	3	3	5	9	21
10	5	20	21	23	35
11	0	266.	1060.	2344.	11279.
12	0	118.	279.	494.	960
13	21	53	59	64	82
14	0	50	66	80	100
15	0	4	5	7	10
16	1	5	6	7	7
17	1	5	6	7	7
18	1	5	6	6	7
19	1	1	2	4	7
20	1	1	2	3	7
21	1	1	2	3	7
22	1	5	5	6	7
23	1	5	6	6	7
24	1	5	5	6	7
25	1	1	2	4	7
26	1	1	2	4	7
27	1	1	2	3	7
28	1	3	4	5	7
29	1	3	5	6	7
30	1	2	4	5	7
31	1	1	2	3	7
32	1	1	2	3	7
33	1	1	1	3	7
34	1	4	4	4	7
35	1	4	4	6	7
36	1	4	4	5	7
37	1	4	5	6	7
38	1	3	4	4	7
39	1	2	3	4	5
40	1	3	3	4	5
41	1	3	4	4	5
42	1	3	3	4	5
43	1	3	4	4	5
44	1	2.75	3	4	5
45	1	3	3	4	5
46	0	3	8	12	14
47	0	17.7	70.7	156.	752.
48	0	0	21.0	126.	825.

49	0	36.6	87.4	149.	450.
50	0	0.679	1.09	1.74	3.87
51	0	13.7	63.7	149.	745.
52	0	8.25	72	184.	847.
53	0	0	0	0	359.
54	0	0.733	49.3	131.	752.
55	0	0	0	0	261.
56	0	0	0	0	661.
57	0	1	1	1	4
58	1	2	2	3	3
59	580	826.	937.	1100.	2267.
60	0	19.5	54.9	111.	1308.
61	-441.	-36.6	-10.0	0	576.
62	21	53	59	64	82
63	101	385	563	862	11124
64	0	0	10	28	30
65	1	5	6	6	6
66	18	24	27	31	40
67	145.	168.	175.	180.	203.
68	31.8	66.2	81.6	97.5	227.
69	30	600	900	1500	3000
70	0	57	86	100	100
71	0	10	32	75	100
72	0	5	15	35	70
73	0	6	19	50	100
74	0	5	11	24	70
75	0	7	15	30	90
76	0	15	40	61	85
77	0	0	70	100	100
78	0	67	91	100	100
79	0	0.353	0.670	0.884	1
80	0	1	1	1	1
81	145.	168.	175.	180.	203.
82	4.54	66	81.6	97.5	227.
83	0	0	0	0	1
84	0	0	0	0	1
85	0	0	0	0	1

```
-- Variable type: POSIXct -----
  skim_variable              n_missing complete_rate min
1 date                      0          1    2025-02-12 19:14:34
2 nintendo_account_creation_date 4251      0.272 2016-02-17 09:28:57
  max                      median          n_unique
```

1	2025-09-24 23:47:26	2025-06-30 14:06:56	5829
2	2025-06-04 09:06:37	2019-02-02 08:07:40	377

```
comp2_nodes <- isolated_components |>
  filter(component == 2) |>
  pull(node)
unique_rows <- unique(unlist(mapper_graph_data$points_in_vertex[c(comp2_nodes)]))
comp2_data <- mapper_full_data[unique_rows, ]
glimpse(comp2_data)
```

```
Rows: 1
Columns: 157
$ wave <dbl> 1
$ aff_val <dbl> 75
$ wemwbs_total <dbl> 22
$ bangs_auto_sat <dbl> 13
$ bangs_auto_frus <dbl> 14
$ bangs_comp_sat <dbl> 17
$ bangs_comp_frus <dbl> 11
$ bangs_rel_sat <dbl> 17
$ bangs_rel_frus <dbl> 10
$ gaming_value_sum <dbl> 26
$ total_wave_minutes <dbl> 234
$ max_binge_minutes <dbl> 154
$ atten_score_mean <dbl> 45
$ pid <chr> "p1212627553"
$ adhd <fct> Neurotypical
$ date <dtm> 2025-06-16 23:40:55
$ affective_valence <dbl> 75
$ life_sat <dbl> 5
$ bangs_1 <dbl> 3
$ bangs_2 <dbl> 5
$ bangs_3 <dbl> 5
$ bangs_4 <dbl> 5
$ bangs_5 <dbl> 6
$ bangs_6 <dbl> 3
$ bangs_7 <dbl> 6
$ bangs_8 <dbl> 6
$ bangs_9 <dbl> 5
$ bangs_10 <dbl> 4
$ bangs_11 <dbl> 3
$ bangs_12 <dbl> 4
```

\$ bangs_13	<dbl> 7
\$ bangs_14	<dbl> 6
\$ bangs_15	<dbl> 4
\$ bangs_16	<dbl> 3
\$ bangs_17	<dbl> 3
\$ bangs_18	<dbl> 4
\$ gaming_value_work	<dbl> 6
\$ gaming_value_social	<dbl> 4
\$ gaming_value_cognitive	<dbl> 5
\$ gaming_value_emotion	<dbl> 6
\$ gaming_value_routines	<dbl> 5
\$ wemwbs_1	<dbl> 3
\$ wemwbs_2	<dbl> 3
\$ wemwbs_3	<dbl> 4
\$ wemwbs_4	<dbl> 4
\$ wemwbs_5	<dbl> 2
\$ wemwbs_6	<dbl> 3
\$ wemwbs_7	<dbl> 3
\$ days_played	<int> 2
\$ mean_daily_minutes	<dbl> 15.6
\$ median_daily_minutes	<dbl> 0
\$ sd_daily_minutes	<dbl> 43.47873
\$ cv_daily_minutes	<dbl> 2.787098
\$ binge_day_of_week	<fct> Tue
\$ mean_weekday_minutes	<dbl> 21.27273
\$ mean_weekend_minutes	<dbl> 0
\$ mean_mobile	<dbl> 0
\$ mean_steam	<dbl> 15.6
\$ mean_nintendo	<dbl> 0
\$ mean_xbox	<dbl> 0
\$ platforms_used	<int> 1
\$ simon_waves_count	<int> 1
\$ rt_mean	<dbl> 1472.867
\$ rt_sd	<dbl> 0
\$ rt_roc	<dbl> 0
\$ score_mean	<dbl> 45
\$ qualified	<lgl> TRUE
\$ survey_duration	<dbl> 625
\$ country	<chr> "UK"
\$ geo_area	<chr> "BR7"
\$ panel_source	<chr> "Prolific"
\$ cohort	<chr> "Platform Prolific"
\$ diaries_completed	<dbl> 0

\$ panels_completed	<dbl> 1
\$ local_timezone	<chr> "GMT+0000"
\$ age	<dbl> 28
\$ gender	<chr> "Man"
\$ height	<dbl> 182.88
\$ weight	<dbl> 83.46
\$ ethnicity	<chr> "White"
\$ edu_level	<chr> "Higher education but no degree"
\$ employment	<chr> "Working part-time"
\$ marital_status	<chr> "Single, never married"
\$ political_party	<chr> "Labour Party / Co-operative Party"
\$ self_reported_weekly_play	<dbl> 600
\$ used_platforms	<chr> "Steam"
\$ eligible_platforms	<chr> "Steam"
\$ linked_platforms	<chr> "Steam"
\$ plays_xbox	<lgl> FALSE
\$ plays_steam	<lgl> TRUE
\$ plays_nintendo	<lgl> FALSE
\$ plays_ios	<lgl> FALSE
\$ plays_android	<lgl> FALSE
\$ plays_playstation	<lgl> FALSE
\$ prop_steam	<dbl> 100
\$ prop_xbox	<dbl> NA
\$ prop_playstation	<dbl> NA
\$ prop_nintendo	<dbl> NA
\$ prop_ios	<dbl> NA
\$ prop_android	<dbl> NA
\$ prop_other	<dbl> NA
\$ nintendo_uses_family	<chr> NA
\$ android_uses_family	<chr> NA
\$ ios_uses_family	<chr> NA
\$ steam_uses_family	<chr> "No"
\$ xbox_uses_family	<chr> NA
\$ total_recent_play_accounted_for	<dbl> 100
\$ total_play_possible_to_collect	<dbl> 100
\$ neuro_identify	<chr> "Yes"
\$ neuro_diagnosed	<chr> "No"
\$ neuro_diag_asd	<chr> NA
\$ neuro_diag_adhd	<chr> NA
\$ neuro_diag_dyslexia	<chr> NA
\$ neuro_diag_dyspraxia	<chr> NA
\$ neuro_diag_dyscalculia	<chr> NA
\$ neuro_diag_tourette	<chr> NA

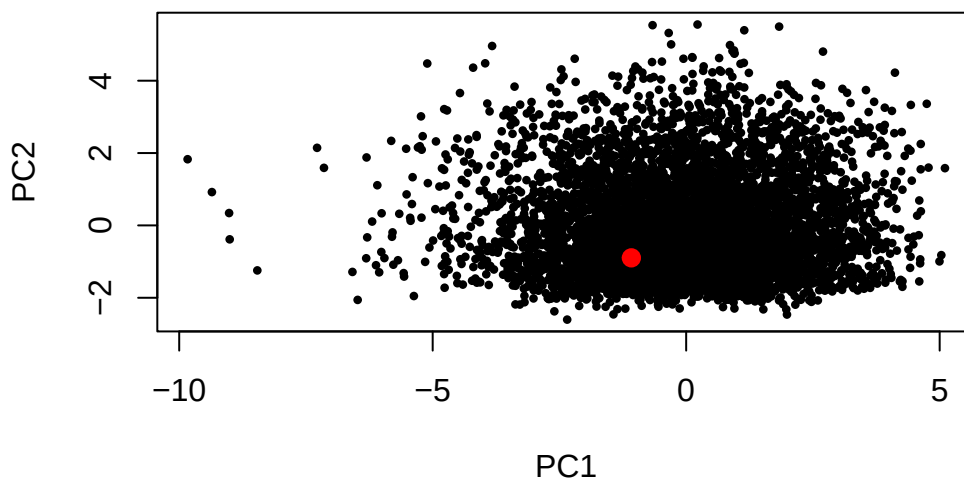
\$ neuro_diag_ocd	<chr> NA
\$ neuro_diag_spd	<chr> NA
\$ neuro_diag_nvld	<chr> NA
\$ neuro_diag_specific_ld	<chr> NA
\$ neuro_diag_prefer_not_say	<chr> NA
\$ neuro_diag_other	<chr> NA
\$ neuro_iden_asd	<chr> "Autism Spectrum Disorder (ASD)"
\$ neuro_iden_adhd	<chr> NA
\$ neuro_iden_dyslexia	<chr> NA
\$ neuro_iden_dyspraxia	<chr> NA
\$ neuro_iden_dyscalculia	<chr> NA
\$ neuro_iden_tourette	<lgl> NA
\$ neuro_iden_ocd	<chr> NA
\$ neuro_iden_spd	<chr> NA
\$ neuro_iden_nvld	<lgl> NA
\$ neuro_iden_specific_ld	<chr> NA
\$ neuro_iden_prefer_not_say	<chr> NA
\$ neuro_iden_other	<chr> NA
\$ care_children	<chr> NA
\$ care_elderly	<chr> NA
\$ care_disability	<chr> NA
\$ care_none	<chr> "No, I do not have additional caregivi~
\$ care_prefer_not_say	<chr> NA
\$ care_other	<chr> NA
\$ dependents	<chr> "None"
\$ nintendo_account_creation_date	<dtm> NA
\$ nintendo_first_party_prop	<dbl> NA
\$ geo_area_original	<chr> "BR7"
\$ gender_original	<chr> "Man"
\$ edu_level_original	<chr> "Some college but no degree"
\$ ethnicity_original	<chr> "White"
\$ marital_status_original	<chr> "Single, never married"
\$ political_party_original	<chr> "Labour Party / Co-operative Party"
\$ employment_original	<chr> "Working part-time"
\$ dependents_original	<chr> NA
\$ dependents_imputed	<dbl> 1
\$ height_original	<dbl> 182.88
\$ weight_original	<dbl> 83.46
\$ height_imputed	<dbl> 0
\$ weight_imputed	<dbl> 0
\$ any_adhd	<dbl> 0

```

plot(
  mapper_pca$x[, 1:2],
  col = "black",
  pch = 16,
  cex = 0.6,
  main = "Component 6 data highlight",
  xlab = "PC1",
  ylab = "PC2"
)
points(
  mapper_pca$x[unique_rows, 1:2, drop = FALSE],
  col = "red",
  pch = 19,
  cex = 1.2
)

```

Component 6 data highlight



The isolated component of 4 nodes consists of only one data point from the biweekly data. They are neurotypical, had their highest binge of wave 1 on a Tuesday. They played very little during the two weeks (234 minutes). As the total wemwbs score for someone saying “some of the time” to the positive questions is 22, they appear to have slightly lower wellbeing, but not noticeably. However, they say that gaming has an influence on their life (gaming_value_sum = 26) and is almost classed as an outlier based on the 1.5IQR rule. Switching to the gaming value colourisation shows this as red. They also show higher BANGS relatedness satisfaction (17), which is significantly higher than average. Despite not playing lots during the time, half their wave telemetry was in one session, and the participant felt a strong connection with the

players/characters around them. As they play steam games, I want to check the metadata for this player, looking into what they generally play...

```
comp3_nodes <- isolated_components |>
  filter(component == 3) |>
  pull(node)
unique_rows <- unique(unlist(mapper_graph_data$points_in_vertex[c(comp3_nodes)]))
comp3_data <- mapper_full_data[unique_rows, ]
glimpse(comp3_data)
```

Rows: 1

Columns: 157

\$ wave	<dbl> 2
\$ aff_val	<dbl> 40
\$ wemwbs_total	<dbl> 12
\$ bangs_auto_sat	<dbl> 15
\$ bangs_auto_frus	<dbl> 20
\$ bangs_comp_sat	<dbl> 20
\$ bangs_comp_frus	<dbl> 13
\$ bangs_rel_sat	<dbl> 16
\$ bangs_rel_frus	<dbl> 21
\$ gaming_value_sum	<dbl> 27
\$ total_wave_minutes	<dbl> 1746.85
\$ max_binge_minutes	<dbl> 521.7167
\$ atten_score_mean	<dbl> 68
\$ pid	<chr> "p1219663417"
\$ adhd	<fct> Diagnosed ADHD
\$ date	<dtm> 2025-04-11 14:12:41
\$ affective_valence	<dbl> 40
\$ life_sat	<dbl> 2
\$ bangs_1	<dbl> 7
\$ bangs_2	<dbl> 7
\$ bangs_3	<dbl> 1
\$ bangs_4	<dbl> 7
\$ bangs_5	<dbl> 6
\$ bangs_6	<dbl> 7
\$ bangs_7	<dbl> 7
\$ bangs_8	<dbl> 6
\$ bangs_9	<dbl> 7
\$ bangs_10	<dbl> 1
\$ bangs_11	<dbl> 7
\$ bangs_12	<dbl> 5

\$ bangs_13	<dbl> 5
\$ bangs_14	<dbl> 6
\$ bangs_15	<dbl> 5
\$ bangs_16	<dbl> 7
\$ bangs_17	<dbl> 7
\$ bangs_18	<dbl> 7
\$ gaming_value_work	<dbl> 4
\$ gaming_value_social	<dbl> 7
\$ gaming_value_cognitive	<dbl> 7
\$ gaming_value_emotion	<dbl> 6
\$ gaming_value_routines	<dbl> 3
\$ wemwbs_1	<dbl> 1
\$ wemwbs_2	<dbl> 2
\$ wemwbs_3	<dbl> 2
\$ wemwbs_4	<dbl> 3
\$ wemwbs_5	<dbl> 1
\$ wemwbs_6	<dbl> 2
\$ wemwbs_7	<dbl> 1
\$ days_played	<int> 9
\$ mean_daily_minutes	<dbl> 116.4567
\$ median_daily_minutes	<dbl> 19
\$ sd_daily_minutes	<dbl> 172.8191
\$ cv_daily_minutes	<dbl> 1.483978
\$ binge_day_of_week	<fct> Thu
\$ mean_weekday_minutes	<dbl> 143.1682
\$ mean_weekend_minutes	<dbl> 43
\$ mean_mobile	<dbl> 0
\$ mean_steam	<dbl> 114.3544
\$ mean_nintendo	<dbl> 2.102222
\$ mean_xbox	<dbl> 0
\$ platforms_used	<int> 2
\$ simon_waves_count	<int> 3
\$ rt_mean	<dbl> 752.7636
\$ rt_sd	<dbl> 80.47057
\$ rt_roc	<dbl> -40.21087
\$ score_mean	<dbl> 68
\$ qualified	<lgl> TRUE
\$ survey_duration	<dbl> 650
\$ country	<chr> "UK"
\$ geo_area	<chr> "LS28"
\$ panel_source	<chr> "Prolific"
\$ cohort	<chr> "Platform Prolific"
\$ diaries_completed	<dbl> 0

\$ panels_completed	<dbl> 4
\$ local_timezone	<chr> "GMT+0000"
\$ age	<dbl> 25
\$ gender	<chr> "Non-binary"
\$ height	<dbl> 182.88
\$ weight	<dbl> 79.83
\$ ethnicity	<chr> "White"
\$ edu_level	<chr> "Higher education but no degree"
\$ employment	<chr> "Not currently employed"
\$ marital_status	<chr> "Single, never married"
\$ political_party	<chr> "Green Party of England and Wales"
\$ self_reported_weekly_play	<dbl> NA
\$ used_platforms	<chr> "Steam,Xbox,Nintendo Switch,Android"
\$ eligible_platforms	<chr> "Xbox, Steam, Nintendo"
\$ linked_platforms	<chr> "Steam, Nintendo"
\$ plays_xbox	<lgl> TRUE
\$ plays_steam	<lgl> TRUE
\$ plays_nintendo	<lgl> TRUE
\$ plays_ios	<lgl> FALSE
\$ plays_android	<lgl> TRUE
\$ plays_playstation	<lgl> FALSE
\$ prop_steam	<dbl> 57
\$ prop_xbox	<dbl> 28
\$ prop_playstation	<dbl> NA
\$ prop_nintendo	<dbl> 5
\$ prop_ios	<dbl> NA
\$ prop_android	<dbl> 10
\$ prop_other	<dbl> NA
\$ nintendo_uses_family	<chr> "No"
\$ android_uses_family	<chr> NA
\$ ios_uses_family	<chr> NA
\$ steam_uses_family	<chr> "No"
\$ xbox_uses_family	<chr> NA
\$ total_recent_play_accounted_for	<dbl> 57
\$ total_play_possible_to_collect	<dbl> 62
\$ neuro_identify	<chr> "Yes"
\$ neuro_diagnosed	<chr> "Yes"
\$ neuro_diag_asd	<chr> "Autism Spectrum Disorder (ASD)"
\$ neuro_diag_adhd	<chr> "Attention Deficit Hyperactivity Disor~
\$ neuro_diag_dyslexia	<chr> NA
\$ neuro_diag_dyspraxia	<chr> NA
\$ neuro_diag_dyscalculia	<chr> NA
\$ neuro_diag_tourette	<chr> NA

```

$ neuro_diag_ocd           <chr> NA
$ neuro_diag_spd           <chr> NA
$ neuro_diag_nvld         <chr> NA
$ neuro_diag_specific_ld   <chr> NA
$ neuro_diag_prefer_not_say <chr> NA
$ neuro_diag_other         <chr> NA
$ neuro_iden_asd           <chr> NA
$ neuro_iden_adhd          <chr> NA
$ neuro_iden_dyslexia      <chr> NA
$ neuro_iden_dyspraxia     <chr> NA
$ neuro_iden_dyscalculia   <chr> NA
$ neuro_iden_tourette      <lgl> NA
$ neuro_iden_ocd           <chr> NA
$ neuro_iden_spd           <chr> NA
$ neuro_iden_nvld         <lgl> NA
$ neuro_iden_specific_ld   <chr> NA
$ neuro_iden_prefer_not_say <chr> NA
$ neuro_iden_other         <chr> NA
$ care_children            <chr> NA
$ care_elderly             <chr> NA
$ care_disability          <chr> NA
$ care_none                <chr> "No, I do not have additional caregivi~
$ care_prefer_not_say      <chr> NA
$ care_other               <chr> NA
$ dependents               <chr> "None"
$ nintendo_account_creation_date <dtm> 2017-02-05 00:19:06
$ nintendo_first_party_prop <dbl> 0.9967589
$ geo_area_original        <chr> "LS28"
$ gender_original          <chr> "Non-binary"
$ edu_level_original       <chr> "Some college but no degree"
$ ethnicity_original       <chr> "White"
$ marital_status_original  <chr> "Single, never married"
$ political_party_original <chr> "Green Party of England and Wales"
$ employment_original      <chr> "Not currently employed"
$ dependents_original      <chr> NA
$ dependents_imputed       <dbl> 1
$ height_original          <dbl> 182.88
$ weight_original          <dbl> 79.83
$ height_imputed           <dbl> 0
$ weight_imputed           <dbl> 0
$ any_adhd                 <dbl> 1

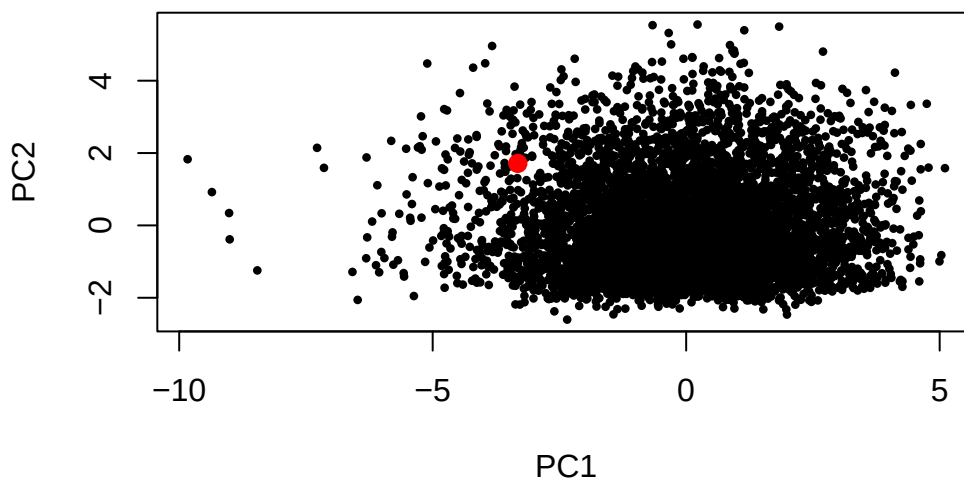
```

```

plot(
  mapper_pca$x[, 1:2],
  col = "black",
  pch = 16,
  cex = 0.6,
  main = "Component 3 data highlight",
  xlab = "PC1",
  ylab = "PC2"
)
points(
  mapper_pca$x[unique_rows, 1:2, drop = FALSE],
  col = "red",
  pch = 19,
  cex = 1.2
)

```

Component 3 data highlight



Another component is a singular node and data point. They are a very interesting case: low wemwbs (12), life satisfaction (2), affective valence (40), very high autonomy frustration (20), high competence satisfaction (20), very high relatedness frustration (21, the maximum value). They played a total of more than 1740 minutes in the two weeks (29 hours), with over 500 minutes in one session. Although only on instance and is not actually a high total telemetry, the long session highlights extreme hyperfocus. They also achieved a higher attention score of 68. The video game type needs to be checked; playing first person shooters may show higher relatedness frustration as players can experience toxic behaviour.

```

comp5_nodes <- isolated_components |>
  filter(component == 5) |>
  pull(node)
unique_rows <- unique(unlist(mapper_graph_data$points_in_vertex[c(comp5_nodes)]))
comp5_data <- mapper_full_data[unique_rows, ]
glimpse(comp5_data)

```

```

Rows: 1
Columns: 157
$ wave                                <dbl> 1
$ aff_val                             <dbl> 20
$ wemwbs_total                        <dbl> 27
$ bangs_auto_sat                     <dbl> 3
$ bangs_auto_frus                    <dbl> 21
$ bangs_comp_sat                     <dbl> 3
$ bangs_comp_frus                    <dbl> 9
$ bangs_rel_sat                      <dbl> 3
$ bangs_rel_frus                     <dbl> 9
$ gaming_value_sum                   <dbl> 20
$ total_wave_minutes                 <dbl> 686.7554
$ max_binge_minutes                  <dbl> 685.7554
$ atten_score_mean                   <dbl> 54
$ pid                               <chr> "p3011712557"
$ adhd                               <fct> Neurotypical
$ date                               <dtm> 2025-04-01 20:56:05
$ affective_valence                  <dbl> 20
$ life_sat                           <dbl> 2
$ bangs_1                            <dbl> 1
$ bangs_2                            <dbl> 1
$ bangs_3                            <dbl> 1
$ bangs_4                            <dbl> 7
$ bangs_5                            <dbl> 7
$ bangs_6                            <dbl> 7
$ bangs_7                            <dbl> 1
$ bangs_8                            <dbl> 1
$ bangs_9                            <dbl> 1
$ bangs_10                           <dbl> 7
$ bangs_11                           <dbl> 1
$ bangs_12                           <dbl> 1
$ bangs_13                           <dbl> 1
$ bangs_14                           <dbl> 1
$ bangs_15                           <dbl> 1

```

\$ bangs_16	<dbl> 7
\$ bangs_17	<dbl> 1
\$ bangs_18	<dbl> 1
\$ gaming_value_work	<dbl> 4
\$ gaming_value_social	<dbl> 4
\$ gaming_value_cognitive	<dbl> 4
\$ gaming_value_emotion	<dbl> 4
\$ gaming_value_routines	<dbl> 4
\$ wemwbs_1	<dbl> 1
\$ wemwbs_2	<dbl> 5
\$ wemwbs_3	<dbl> 5
\$ wemwbs_4	<dbl> 1
\$ wemwbs_5	<dbl> 5
\$ wemwbs_6	<dbl> 5
\$ wemwbs_7	<dbl> 5
\$ days_played	<int> 2
\$ mean_daily_minutes	<dbl> 45.78369
\$ median_daily_minutes	<dbl> 0
\$ sd_daily_minutes	<dbl> 177.043
\$ cv_daily_minutes	<dbl> 3.866945
\$ binge_day_of_week	<fct> Sat
\$ mean_weekday_minutes	<dbl> 0.09090909
\$ mean_weekend_minutes	<dbl> 171.4388
\$ mean_mobile	<dbl> 0
\$ mean_steam	<dbl> 45.78369
\$ mean_nintendo	<dbl> 0
\$ mean_xbox	<dbl> 0
\$ platforms_used	<int> 1
\$ simon_waves_count	<int> 2
\$ rt_mean	<dbl> 1056.306
\$ rt_sd	<dbl> 270.8742
\$ rt_roc	<dbl> -191.537
\$ score_mean	<dbl> 54
\$ qualified	<lgl> TRUE
\$ survey_duration	<dbl> 403
\$ country	<chr> "US"
\$ geo_area	<chr> "182"
\$ panel_source	<chr> "Prolific"
\$ cohort	<chr> "Platform Prolific"
\$ diaries_completed	<dbl> 29
\$ panels_completed	<dbl> 5
\$ local_timezone	<chr> "GMT-0500"
\$ age	<dbl> 22

\$ gender	<chr> "Woman"
\$ height	<dbl> 167.64
\$ weight	<dbl> 108.86
\$ ethnicity	<chr> "White"
\$ edu_level	<chr> "Higher education but no degree"
\$ employment	<chr> "Not currently employed"
\$ marital_status	<chr> "In a domestic partnership or civil un~"
\$ political_party	<chr> "Independent"
\$ self_reported_weekly_play	<dbl> 180
\$ used_platforms	<chr> "Steam"
\$ eligible_platforms	<chr> "Steam"
\$ linked_platforms	<chr> "Steam"
\$ plays_xbox	<lgl> FALSE
\$ plays_steam	<lgl> TRUE
\$ plays_nintendo	<lgl> FALSE
\$ plays_ios	<lgl> FALSE
\$ plays_android	<lgl> FALSE
\$ plays_playstation	<lgl> FALSE
\$ prop_steam	<dbl> 100
\$ prop_xbox	<dbl> NA
\$ prop_playstation	<dbl> NA
\$ prop_nintendo	<dbl> NA
\$ prop_ios	<dbl> NA
\$ prop_android	<dbl> NA
\$ prop_other	<dbl> NA
\$ nintendo_uses_family	<chr> NA
\$ android_uses_family	<chr> NA
\$ ios_uses_family	<chr> NA
\$ steam_uses_family	<chr> "No"
\$ xbox_uses_family	<chr> NA
\$ total_recent_play_accounted_for	<dbl> 100
\$ total_play_possible_to_collect	<dbl> 100
\$ neuro_identify	<chr> "No"
\$ neuro_diagnosed	<chr> NA
\$ neuro_diag_asd	<chr> NA
\$ neuro_diag_adhd	<chr> NA
\$ neuro_diag_dyslexia	<chr> NA
\$ neuro_diag_dyspraxia	<chr> NA
\$ neuro_diag_dyscalculia	<chr> NA
\$ neuro_diag_tourette	<chr> NA
\$ neuro_diag OCD	<chr> NA
\$ neuro_diag_spd	<chr> NA
\$ neuro_diag_nvld	<chr> NA


```

$ neuro_diag_specific_ld      <chr> NA
$ neuro_diag_prefer_not_say   <chr> NA
$ neuro_diag_other            <chr> NA
$ neuro_iden_asd              <chr> NA
$ neuro_iden_adhd             <chr> NA
$ neuro_iden_dyslexia         <chr> NA
$ neuro_iden_dyspraxia        <chr> NA
$ neuro_iden_dyscalculia      <chr> NA
$ neuro_iden_tourette         <lgl> NA
$ neuro_iden OCD              <chr> NA
$ neuro_iden SPD              <chr> NA
$ neuro_iden NVLD             <lgl> NA
$ neuro_iden_specific_ld     <chr> NA
$ neuro_iden_prefer_not_say   <chr> NA
$ neuro_iden_other            <chr> NA
$ care_children               <chr> NA
$ care_elderly                <chr> NA
$ care_disability             <chr> "Yes, I care for a family member with ~
$ care_none                   <chr> NA
$ care_prefer_not_say         <chr> NA
$ care_other                  <chr> NA
$ dependents                  <chr> "1"
$ nintendo_account_creation_date <dtm> NA
$ nintendo_first_party_prop   <dbl> NA
$ geo_area_original           <chr> "182"
$ gender_original             <chr> "Woman"
$ edu_level_original          <chr> "Some college but no degree"
$ ethnicity_original          <chr> "White alone"
$ marital_status_original     <chr> "In a domestic partnership or civil un~
$ political_party_original    <chr> "Independent"
$ employment_original         <chr> "Not currently employed"
$ dependents_original         <chr> "1"
$ dependents_imputed          <dbl> 0
$ height_original             <dbl> 167.64
$ weight_original             <dbl> 108.86
$ height_imputed              <dbl> 0
$ weight_imputed              <dbl> 0
$ any_adhd                    <dbl> 0

```

```

plot(
  mapper_pca$x[, 1:2],
  col = "black",

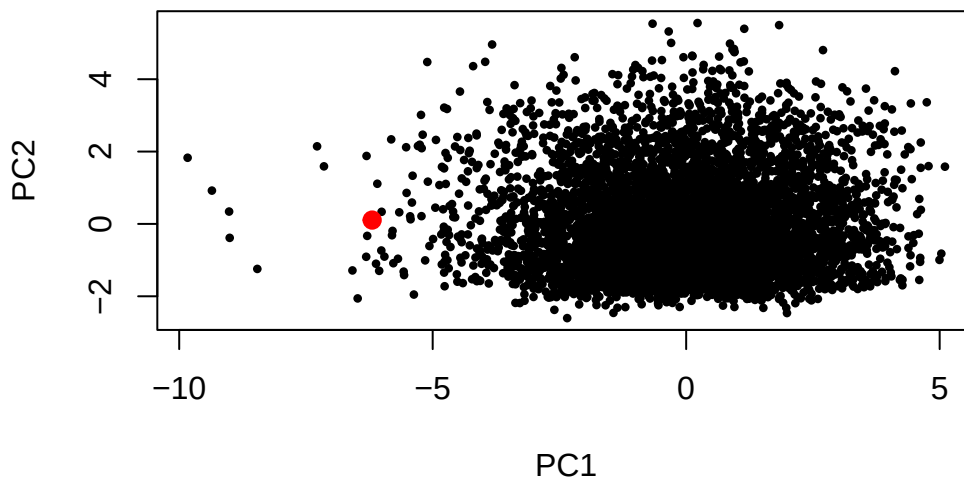
```

```

pch = 16,
cex = 0.6,
main = "Component 5 data highlight",
xlab = "PC1",
ylab = "PC2"
)
points(
  mapper_pca$x[unique_rows, 1:2, drop = FALSE],
  col = "red",
  pch = 19,
  cex = 1.2
)

```

Component 5 data highlight



There is another singular node and data point. They are a neurotypical participant and with their total telemetry (687) almost being their highest binge session (686) on a saturday. This also exhibits hyperfocus, an ADHD trait. But with very few sessions it does not mean they secretly have ADHD. They are the 75th percentile in wemwbs (high wellbeing) but have a very low affective valence (20) and life satisfaction (2). This is odd, and could be an input error. Gaming does not have a large influence on their social life. They experience very low satisfactions (3, 3, 3 for all) but also lower frustrations (9, 9 for comp rel). Interestingly, they have exceedingly high autonomy frustration. With such a large frustration, why do they have a large binge session of more than 11 hours? I need to check the game they played, and why the inputs to the survey are odd (just filling it out weirdly, although they didn't fail the check).

```

comp4_nodes <- isolated_components |>
  filter(component == 4) |>
  pull(node)
unique_rows <- unique(unlist(mapper_graph_data$points_in_vertex[c(comp4_nodes)]))
comp4_data <- mapper_full_data[unique_rows, ]
glimpse(comp4_data)

```

Rows: 2

Columns: 157

```

$ wave                <dbl> 2, 3
$ aff_val              <dbl> 40, 5
$ wemwbs_total        <dbl> 20, 10
$ bangs_auto_sat      <dbl> 11, 4
$ bangs_auto_frus     <dbl> 13, 21
$ bangs_comp_sat      <dbl> 9, 3
$ bangs_comp_frus     <dbl> 11, 16
$ bangs_rel_sat       <dbl> 10, 3
$ bangs_rel_frus      <dbl> 18, 21
$ gaming_value_sum    <dbl> 15, 20
$ total_wave_minutes  <dbl> 5007.1000, 477.4833
$ max_binge_minutes   <dbl> 622.7811, 264.3833
$ atten_score_mean    <dbl> 62.33333, 68.00000
$ pid                 <chr> "p7842684564", "p1219663417"
$ adhd                <fct> Neurotypical, Diagnosed ADHD
$ date                <dtm> 2025-04-12 17:44:26, 2025-04-25 13:17:~
$ affective_valence   <dbl> 40, 5
$ life_sat            <dbl> 3, 0
$ bangs_1             <dbl> 2, 1
$ bangs_2             <dbl> 6, 1
$ bangs_3             <dbl> 3, 2
$ bangs_4             <dbl> 1, 7
$ bangs_5             <dbl> 5, 7
$ bangs_6             <dbl> 7, 7
$ bangs_7             <dbl> 5, 1
$ bangs_8             <dbl> 2, 1
$ bangs_9             <dbl> 2, 1
$ bangs_10            <dbl> 5, 5
$ bangs_11            <dbl> 2, 6
$ bangs_12            <dbl> 4, 5
$ bangs_13            <dbl> 3, 1
$ bangs_14            <dbl> 5, 1
$ bangs_15            <dbl> 2, 1

```

\$ bangs_16	<dbl> 6, 7
\$ bangs_17	<dbl> 6, 7
\$ bangs_18	<dbl> 6, 7
\$ gaming_value_work	<dbl> 2, 4
\$ gaming_value_social	<dbl> 5, 4
\$ gaming_value_cognitive	<dbl> 5, 4
\$ gaming_value_emotion	<dbl> 2, 4
\$ gaming_value_routines	<dbl> 1, 4
\$ wemwbs_1	<dbl> 3, 1
\$ wemwbs_2	<dbl> 3, 1
\$ wemwbs_3	<dbl> 3, 3
\$ wemwbs_4	<dbl> 4, 1
\$ wemwbs_5	<dbl> 1, 2
\$ wemwbs_6	<dbl> 2, 1
\$ wemwbs_7	<dbl> 4, 1
\$ days_played	<int> 12, 5
\$ mean_daily_minutes	<dbl> 333.80667, 31.83222
\$ median_daily_minutes	<dbl> 355.1819, 0.0000
\$ sd_daily_minutes	<dbl> 210.72445, 72.03025
\$ cv_daily_minutes	<dbl> 0.6312769, 2.2628094
\$ binge_day_of_week	<fct> Fri, Thu
\$ mean_weekday_minutes	<dbl> 388.42880, 32.77121
\$ mean_weekend_minutes	<dbl> 224.5624, 29.2500
\$ mean_mobile	<dbl> 0, 0
\$ mean_steam	<dbl> 333.80667, 31.55889
\$ mean_nintendo	<dbl> 0.0000000, 0.2733333
\$ mean_xbox	<dbl> 0, 0
\$ platforms_used	<int> 1, 2
\$ simon_waves_count	<int> 3, 3
\$ rt_mean	<dbl> 858.3859, 752.7636
\$ rt_sd	<dbl> 45.48507, 80.47057
\$ rt_roc	<dbl> 8.05018, -40.21087
\$ score_mean	<dbl> 62.33333, 68.00000
\$ qualified	<lgl> TRUE, TRUE
\$ survey_duration	<dbl> 607, 650
\$ country	<chr> "UK", "UK"
\$ geo_area	<chr> "M23", "LS28"
\$ panel_source	<chr> "Prolific", "Prolific"
\$ cohort	<chr> "Platform Prolific", "Platform Prolifi~
\$ diaries_completed	<dbl> 0, 0
\$ panels_completed	<dbl> 6, 4
\$ local_timezone	<chr> "GMT+0000", "GMT+0000"
\$ age	<dbl> 23, 25

\$ gender	<chr> "Woman", "Non-binary"
\$ height	<dbl> 165.10, 182.88
\$ weight	<dbl> 108.86, 79.83
\$ ethnicity	<chr> "Asian", "White"
\$ edu_level	<chr> "Higher education but no degree", "Hig~
\$ employment	<chr> "Working part-time", "Not currently em~
\$ marital_status	<chr> "Single, never married", "Single, neve~
\$ political_party	<chr> "Green Party of England and Wales", "G~
\$ self_reported_weekly_play	<dbl> 900, NA
\$ used_platforms	<chr> "Steam,Xbox,iOS", "Steam,Xbox,Nintendo~
\$ eligible_platforms	<chr> "Xbox, Steam", "Xbox, Steam, Nintendo"
\$ linked_platforms	<chr> "Steam", "Steam, Nintendo"
\$ plays_xbox	<lgl> TRUE, TRUE
\$ plays_steam	<lgl> TRUE, TRUE
\$ plays_nintendo	<lgl> FALSE, TRUE
\$ plays_ios	<lgl> TRUE, FALSE
\$ plays_android	<lgl> FALSE, TRUE
\$ plays_playstation	<lgl> FALSE, FALSE
\$ prop_steam	<dbl> 35, 57
\$ prop_xbox	<dbl> 55, 28
\$ prop_playstation	<dbl> NA, NA
\$ prop_nintendo	<dbl> NA, 5
\$ prop_ios	<dbl> 10, NA
\$ prop_android	<dbl> NA, 10
\$ prop_other	<dbl> NA, NA
\$ nintendo_uses_family	<chr> NA, "No"
\$ android_uses_family	<chr> NA, NA
\$ ios_uses_family	<chr> NA, NA
\$ steam_uses_family	<chr> "No", "No"
\$ xbox_uses_family	<chr> NA, NA
\$ total_recent_play_accounted_for	<dbl> 0, 57
\$ total_play_possible_to_collect	<dbl> 35, 62
\$ neuro_identify	<chr> "Prefer not to say", "Yes"
\$ neuro_diagnosed	<chr> NA, "Yes"
\$ neuro_diag_asd	<chr> NA, "Autism Spectrum Disorder (ASD)"
\$ neuro_diag_adhd	<chr> NA, "Attention Deficit Hyperactivity D~
\$ neuro_diag_dyslexia	<chr> NA, NA
\$ neuro_diag_dyspraxia	<chr> NA, NA
\$ neuro_diag_dyscalculia	<chr> NA, NA
\$ neuro_diag_tourette	<chr> NA, NA
\$ neuro_diag OCD	<chr> NA, NA
\$ neuro_diag_spd	<chr> NA, NA
\$ neuro_diag_nvld	<chr> NA, NA

```

$ neuro_diag_specific_ld      <chr> NA, NA
$ neuro_diag_prefer_not_say   <chr> NA, NA
$ neuro_diag_other            <chr> NA, NA
$ neuro_iden_asd              <chr> NA, NA
$ neuro_iden_adhd             <chr> NA, NA
$ neuro_iden_dyslexia         <chr> NA, NA
$ neuro_iden_dyspraxia        <chr> NA, NA
$ neuro_iden_dyscalculia      <chr> NA, NA
$ neuro_iden_tourette         <lgl> NA, NA
$ neuro_iden OCD              <chr> NA, NA
$ neuro_iden_spd              <chr> NA, NA
$ neuro_iden_nvld             <lgl> NA, NA
$ neuro_iden_specific_ld      <chr> NA, NA
$ neuro_iden_prefer_not_say   <chr> NA, NA
$ neuro_iden_other            <chr> NA, NA
$ care_children               <chr> NA, NA
$ care_elderly                <chr> NA, NA
$ care_disability             <chr> NA, NA
$ care_none                   <chr> "No, I do not have additional caregivi~
$ care_prefer_not_say         <chr> NA, NA
$ care_other                  <chr> NA, NA
$ dependents                  <chr> "None", "None"
$ nintendo_account_creation_date <dtm> NA, 2017-02-05 00:19:06
$ nintendo_first_party_prop   <dbl> NA, 0.9967589
$ geo_area_original           <chr> "m23", "LS28"
$ gender_original              <chr> "Woman", "Non-binary"
$ edu_level_original           <chr> "Some college but no degree", "Some co~
$ ethnicity_original           <chr> "Asian or Asian British", "White"
$ marital_status_original      <chr> "Single, never married", "Single, neve~
$ political_party_original     <chr> "Green Party of England and Wales", "G~
$ employment_original         <chr> "Working part-time", "Not currently em~
$ dependents_original         <chr> NA, NA
$ dependents_imputed          <dbl> 1, 1
$ height_original              <dbl> 165.10, 182.88
$ weight_original              <dbl> 108.86, 79.83
$ height_imputed               <dbl> 0, 0
$ weight_imputed               <dbl> 0, 0
$ any_adhd                     <dbl> 0, 1

```

```

plot(
  mapper_pca$x[, 1:2],
  col = "black",

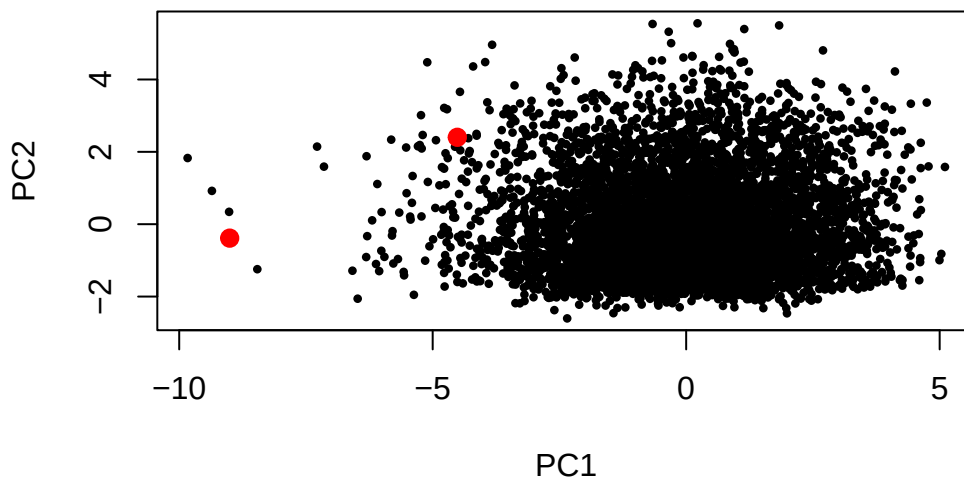
```

```

pch = 16,
cex = 0.6,
main = "Component 4 data highlight",
xlab = "PC1",
ylab = "PC2"
)
points(
  mapper_pca$x[unique_rows, 1:2, drop = FALSE],
  col = "red",
  pch = 19,
  cex = 1.2
)

```

Component 4 data highlight



An isolated component with 3 nodes and 2 rows of data (from two different participants) contains one diagnosed ADHD participant. They are also from two different waves (2 and 3 diagnosed) and have differing wellbeings (20, 10 diagnosed). Diagnosed experienced extreme autonomy frustration, and very low satisfaction, but played a lot less than the neurotypical (477 total compared to 5007). However, their life satisfaction is low (3, 0 diagnosed) and their affective valence (40, 5 diagnosed).

```

comp6_nodes <- isolated_components |>
  filter(component == 6) |>
  pull(node)
unique_rows <- unique(unlist(mapper_graph_data$points_in_vertex[c(comp6_nodes)]))
comp6_data <- mapper_full_data[unique_rows, ]
glimpse(comp6_data)

```

```

Rows: 3
Columns: 157
$ wave <dbl> 2, 6, 2
$ aff_val <dbl> 40, 59, 80
$ wemwbs_total <dbl> 16, 19, 23
$ bangs_auto_sat <dbl> 9, 17, 17
$ bangs_auto_frus <dbl> 15, 12, 12
$ bangs_comp_sat <dbl> 11, 16, 15
$ bangs_comp_frus <dbl> 11, 8, 10
$ bangs_rel_sat <dbl> 14, 11, 5
$ bangs_rel_frus <dbl> 11, 6, 17
$ gaming_value_sum <dbl> 13, 22, 20
$ total_wave_minutes <dbl> 1830.117, 1787.100, 531.550
$ max_binge_minutes <dbl> 520.0000, 326.3333, 363.5500
$ atten_score_mean <dbl> 59.66667, 63.33333, 55.00000
$ pid <chr> "p9775353241", "p7790234409", "p960429~
$ adhd <fct> Self-Identified ADHD, Neurotypical, Ne~
$ date <dtm> 2025-06-30 13:18:42, 2025-09-15 20:35:~
$ affective_valence <dbl> 40, 59, 80
$ life_sat <dbl> 4, 5, 4
$ bangs_1 <dbl> 3, 6, 5
$ bangs_2 <dbl> 3, 5, 6
$ bangs_3 <dbl> 3, 6, 6
$ bangs_4 <dbl> 5, 4, 6
$ bangs_5 <dbl> 5, 4, 2
$ bangs_6 <dbl> 5, 4, 4
$ bangs_7 <dbl> 4, 5, 5
$ bangs_8 <dbl> 4, 6, 6
$ bangs_9 <dbl> 3, 5, 4
$ bangs_10 <dbl> 4, 3, 4
$ bangs_11 <dbl> 3, 2, 3
$ bangs_12 <dbl> 4, 3, 3
$ bangs_13 <dbl> 4, 4, 1
$ bangs_14 <dbl> 5, 3, 1
$ bangs_15 <dbl> 5, 4, 3
$ bangs_16 <dbl> 4, 2, 6
$ bangs_17 <dbl> 5, 2, 6
$ bangs_18 <dbl> 2, 2, 5
$ gaming_value_work <dbl> 1, 4, 3
$ gaming_value_social <dbl> 3, 4, 4
$ gaming_value_cognitive <dbl> 3, 5, 5
$ gaming_value_emotion <dbl> 3, 5, 4
$ gaming_value_routines <dbl> 3, 4, 4

```


\$ wemwbs_1	<dbl> 2, 3, 3
\$ wemwbs_2	<dbl> 2, 3, 4
\$ wemwbs_3	<dbl> 2, 3, 4
\$ wemwbs_4	<dbl> 4, 3, 2
\$ wemwbs_5	<dbl> 2, 3, 3
\$ wemwbs_6	<dbl> 2, 2, 3
\$ wemwbs_7	<dbl> 2, 2, 4
\$ days_played	<int> 11, 11, 3
\$ mean_daily_minutes	<dbl> 122.00778, 119.14000, 35.43667
\$ median_daily_minutes	<dbl> 93.0000, 111.5404, 0.0000
\$ sd_daily_minutes	<dbl> 140.37594, 100.99991, 96.54246
\$ cv_daily_minutes	<dbl> 1.1505491, 0.8477414, 2.7243662
\$ binge_day_of_week	<fct> Sun, Sat, Thu
\$ mean_weekday_minutes	<dbl> 73.05455, 118.60152, 48.32273
\$ mean_weekend_minutes	<dbl> 256.6292, 120.6208, 0.0000
\$ mean_mobile	<dbl> 0, 0, 0
\$ mean_steam	<dbl> 122.00778, 119.14000, 35.43667
\$ mean_nintendo	<dbl> 0, 0, 0
\$ mean_xbox	<dbl> 0, 0, 0
\$ platforms_used	<int> 1, 1, 1
\$ simon_waves_count	<int> 3, 3, 1
\$ rt_mean	<dbl> 884.9576, 910.3738, 960.6230
\$ rt_sd	<dbl> 66.7949, 214.6934, 0.0000
\$ rt_roc	<dbl> 32.18559, -96.55847, 0.00000
\$ score_mean	<dbl> 59.66667, 63.33333, 55.00000
\$ qualified	<lgl> TRUE, TRUE, TRUE
\$ survey_duration	<dbl> 442, 298, 210
\$ country	<chr> "UK", "US", "US"
\$ geo_area	<chr> "EH7", "597", "081"
\$ panel_source	<chr> "Prolific", "Prolific", "Prolific"
\$ cohort	<chr> "Platform Prolific", "Platform Prolifi~
\$ diaries_completed	<dbl> 0, 8, 3
\$ panels_completed	<dbl> 5, 6, 5
\$ local_timezone	<chr> "GMT+0000", "GMT-0700", "GMT-0500"
\$ age	<dbl> 26, 24, 28
\$ gender	<chr> "Man", "Man", "Man"
\$ height	<dbl> 180.34, 200.66, 182.88
\$ weight	<dbl> 72.00, 90.72, 70.31
\$ ethnicity	<chr> "Black", "White", "Asian"
\$ edu_level	<chr> "Associate degree", "Bachelors degree"~
\$ employment	<chr> "Student", "Student", "Not currently e~
\$ marital_status	<chr> "Single, never married", "Single, neve~
\$ political_party	<chr> "Green Party of England and Wales", "I~

\$ self_reported_weekly_play	<dbl> 3000, 1680, 3000
\$ used_platforms	<chr> "Steam", "Steam", "Steam"
\$ eligible_platforms	<chr> "Steam", "Steam", "Steam"
\$ linked_platforms	<chr> "Steam", "Steam", "Steam"
\$ plays_xbox	<lgl> FALSE, FALSE, FALSE
\$ plays_steam	<lgl> TRUE, TRUE, TRUE
\$ plays_nintendo	<lgl> FALSE, FALSE, FALSE
\$ plays_ios	<lgl> FALSE, FALSE, FALSE
\$ plays_android	<lgl> FALSE, FALSE, FALSE
\$ plays_playstation	<lgl> FALSE, FALSE, FALSE
\$ prop_steam	<dbl> 100, 100, 100
\$ prop_xbox	<dbl> NA, NA, NA
\$ prop_playstation	<dbl> NA, NA, NA
\$ prop_nintendo	<dbl> NA, NA, NA
\$ prop_ios	<dbl> NA, NA, NA
\$ prop_android	<dbl> NA, NA, NA
\$ prop_other	<dbl> NA, NA, NA
\$ nintendo_uses_family	<chr> NA, NA, NA
\$ android_uses_family	<chr> NA, NA, NA
\$ ios_uses_family	<chr> NA, NA, NA
\$ steam_uses_family	<chr> "No", "No", "No"
\$ xbox_uses_family	<chr> NA, NA, NA
\$ total_recent_play_accounted_for	<dbl> 100, 100, 100
\$ total_play_possible_to_collect	<dbl> 100, 100, 100
\$ neuro_identify	<chr> "Yes", "No", "No"
\$ neuro_diagnosed	<chr> "No", NA, NA
\$ neuro_diag_asd	<chr> NA, NA, NA
\$ neuro_diag_adhd	<chr> NA, NA, NA
\$ neuro_diag_dyslexia	<chr> NA, NA, NA
\$ neuro_diag_dyspraxia	<chr> NA, NA, NA
\$ neuro_diag_dyscalculia	<chr> NA, NA, NA
\$ neuro_diag_tourette	<chr> NA, NA, NA
\$ neuro_diag OCD	<chr> NA, NA, NA
\$ neuro_diag_spd	<chr> NA, NA, NA
\$ neuro_diag_nvld	<chr> NA, NA, NA
\$ neuro_diag_specific_ld	<chr> NA, NA, NA
\$ neuro_diag_prefer_not_say	<chr> NA, NA, NA
\$ neuro_diag_other	<chr> NA, NA, NA
\$ neuro_iden_asd	<chr> NA, NA, NA
\$ neuro_iden_adhd	<chr> "Attention Deficit Hyperactivity Disor~
\$ neuro_iden_dyslexia	<chr> NA, NA, NA
\$ neuro_iden_dyspraxia	<chr> NA, NA, NA
\$ neuro_iden_dyscalculia	<chr> NA, NA, NA

```

$ neuro_iden_tourette      <lgl> NA, NA, NA
$ neuro_iden_ocd           <chr> NA, NA, NA
$ neuro_iden_spd           <chr> NA, NA, NA
$ neuro_iden_nvld         <lgl> NA, NA, NA
$ neuro_iden_specific_ld   <chr> NA, NA, NA
$ neuro_iden_prefer_not_say <chr> NA, NA, NA
$ neuro_iden_other         <chr> NA, NA, NA
$ care_children            <chr> NA, NA, NA
$ care_elderly             <chr> NA, NA, NA
$ care_disability          <chr> NA, NA, NA
$ care_none                <chr> "No, I do not have additional caregivi~
$ care_prefer_not_say      <chr> NA, NA, NA
$ care_other               <chr> NA, NA, NA
$ dependents               <chr> "None", "None", "None"
$ nintendo_account_creation_date <dtm> NA, NA, NA
$ nintendo_first_party_prop <dbl> NA, NA, NA
$ geo_area_original        <chr> "EH7", "597", "081"
$ gender_original          <chr> "Man", "Man", "Man"
$ edu_level_original       <chr> "Associate degree", "Bachelor's degree~
$ ethnicity_original       <chr> "Black, African, Caribbean or Black Br~
$ marital_status_original  <chr> "Single, never married", "Single, neve~
$ political_party_original <chr> "Green Party of England and Wales", "I~
$ employment_original     <chr> "Student", "Student", "Not currently e~
$ dependents_original      <chr> NA, NA, NA
$ dependents_imputed       <dbl> 1, 1, 1
$ height_original          <dbl> 180.34, 200.66, 182.88
$ weight_original          <dbl> 72.00, 90.72, 70.31
$ height_imputed           <dbl> 0, 0, 0
$ weight_imputed           <dbl> 0, 0, 0
$ any_adhd                 <dbl> 1, 0, 0

```

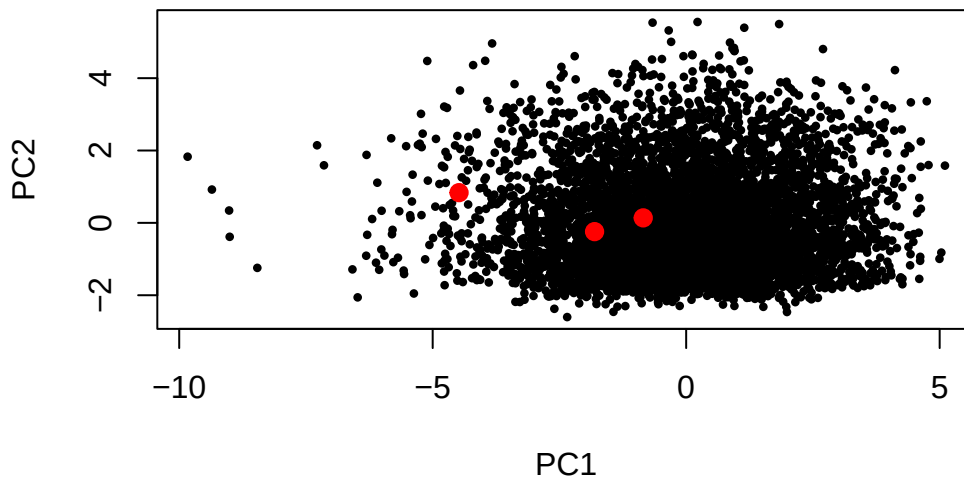
```

plot(
  mapper_pca$x[, 1:2],
  col = "black",
  pch = 16,
  cex = 0.6,
  main = "Component 6 data highlight",
  xlab = "PC1",
  ylab = "PC2"
)
points(
  mapper_pca$x[unique_rows, 1:2, drop = FALSE],

```

```
col = "red",  
pch = 19,  
cex = 1.2  
)
```

Component 6 data highlight



Another component contains just 2 nodes but three participants (one is self-identified ADHD). Two are from the second wave, and one from the sixth. They have varied affective valence (40, 59, 80) and BANGS. However, they have consistent attention scores (the self-identified participant has the attention score in the middle). Two of the total wave minutes are similar (1830, 1787) but one is significantly lower (363). I cannot find a pattern between the three data points, so I wonder if this is the limitation of using PCA with limited variance. Looking at the rotations, I don't see any reason for these data points to be clustered together.